

Renaldo Miya

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EDUCATION

Bachelor of Mechanical Engineering

University of Waterloo

Expected Graduation: April 2029

Waterloo, ON

WORK EXPERIENCE

ARIEL Engineering Assistant

TRIUMF

January 2026 – April 2026

Vancouver, BC

- Designed a compact stainless-steel support frame in SolidWorks for a radioactive gas pumping manifold, forming a rigid, wall-mounted enclosure with minimal floor footprint
- Sized members and clamp layout through iterative ANSYS FEA, holding deflection under 2 mm and stress under 105 MPa while minimizing part count in a radiation-exposed assembly
- Positioned supports and clamps for full technician access to gauges, sensors, and KF connections, enabling independent component removal during maintenance
- Produced ASME Y14.5 GD&T manufacturing drawings for fabrication and inspection, reducing fastener count and streamlining assembly in the ARIEL Target Hall
- Generated and managed BOMs in SolidWorks PDM, coordinating with engineers and technicians to maintain configuration control and design traceability across frame revisions
- Supported fabrication on manual machining tools, validating tolerances and resolving fit-up issues to translate CAD design intent into the installed structure

Mechanical Engineering Co-op

BC Hydro

May 2024 – August 2024

Fort St John, BC

- Reviewed and redlined P&IDs and mechanical layouts against applicable codes and standards, flagging constructability issues and issuing modifications adopted into the build
- Conducted on-site field inspections to verify installation accuracy and code compliance, documenting discrepancies and driving corrective design changes that reduced rework
- Resolved RFIs and Field Change Forms by coordinating across mechanical, electrical, and civil teams, issuing design revisions that kept construction on schedule

Mechanical Design & Composites Member

UW Formula Electric

September 2023 – December 2023

Waterloo, ON

- Designed a relay enclosure in SolidWorks, cutting mass by 25% and reducing part count to speed integration into the vehicle
- Manufactured the composite undertray end-to-end — Mold preparation, body filling, priming, and sanding — achieving a high-quality surface finish and structural integrity

PROJECTS

Robotic Claw | SolidWorks, GD&T, Mechanism Design

Fall 2025

- Designed a motor-driven rack-and-pinion robotic claw that grasps and lifts up to 5 kg, owning it from concept through functional prototype
- Modelled complete assemblies and components in SolidWorks, converting rotational motor output into linear jaw travel through a custom rack-and-pinion drive
- Integrated 3D-printed structural components with machined aluminium parts, balancing weight, manufacturability, and structural stiffness to prevent deflection under peak load
- Designed for serviceability, enabling quick access and independent replacement of motors, fasteners, and power-transmission components

TECHNICAL SKILLS

CAD & Modelling | SolidWorks, AutoCAD

Analysis & Simulation | ANSYS, SolidWorks Simulations, MATLAB

Manufacturing | Mill, Lathe, Drill Press, 3D Printing, Composite Layup

Design & Documentation | GD&T (ASME Y14.5), DFM/DFA, BOM Management, SolidWorks PDM, P&ID

Programming | C++, C, Python, HTML/CSS